

HUNTSVILLE INTL, AL, USA

WMO: 723230

Lat: 34.644N

Lon: 86.786W

Elev: 190

StdP: 99.06

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03856

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.5	(c) -5.2	(d) -14.0	(e) 1.1	(f) -4.6	(g) -11.7	(h) 1.4	(i) -3.2	(j) 10.7	(k) 11.0	(l) 9.3	(m) 9.1	(n) 3.3	(o) 10	(p) 0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	35.2	23.8	34.0	23.7	32.9	23.4	25.9	31.2	25.4	30.7	24.9	30.0	3.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.0	28.1	24.0	19.3	27.6	23.4	18.6	27.1	81.0	31.3	78.5	30.7	76.2	30.1	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 8.0	(c) 7.1	(e) -11.1	(f) 36.8	(g) 2.8	(h) 1.8	(i) -13.1	(j) 38.1	(k) -14.8	(l) 39.1	(m) -16.3	(n) 40.1	(o) -18.4	(p) 41.4		
(a) 9.0	(b) 8.0	(c) 7.1	(e) -12.0	(f) 26.9	(g) 2.7	(h) 0.6	(i) -14.0	(j) 27.4	(k) -15.5	(l) 27.8	(m) -17.0	(n) 28.1	(o) -19.0	(p) 28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	5.3	7.7	11.9	16.9	21.4	25.4	26.7	26.4	23.4	17.3	11.1	7.2
	DBStd	8.78	6.11	5.65	5.32	4.35	3.65	2.56	1.99	2.25	3.46	4.54	4.89	5.43
	HDD10.0	489	171	102	43	4	0	0	0	0	0	3	44	122
	HDD18.3	1649	403	299	206	78	16	0	0	0	6	74	222	346
	CDD10.0	2964	26	37	102	209	355	463	517	508	402	231	77	36
	CDD18.3	1083	1	1	7	33	112	213	259	250	158	43	4	1
	CDH23.3	10122	0	4	58	296	1000	2068	2502	2414	1399	354	25	2
	CDH26.7	4235	0	0	7	61	351	921	1118	1108	579	88	3	0
Wind	WSAvg	3.0	3.6	3.7	3.8	3.6	3.1	2.5	2.1	2.1	2.4	2.7	3.0	3.4
Precipitation	PrecAvg	1424	129	122	164	118	133	108	117	87	102	80	120	146
	PrecMax	1869	277	243	432	319	302	381	376	249	248	306	293	474
	PrecMin	1062	34	15	41	11	33	4	20	15	12	0	16	19
	PrecStd	227	62	62	83	71	63	69	65	53	62	58	55	92
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.8	27.4	30.3	33.3	36.0	36.3	37.5	35.5	31.3	26.3	22.6
		MCWB	16.9	17.2	17.8	19.4	22.2	23.5	24.2	24.0	22.1	21.2	18.8	17.8
	2%	DB	19.0	21.2	24.9	28.5	31.7	34.4	34.7	35.2	33.6	28.9	23.3	20.2
		MCWB	15.8	16.4	16.8	18.7	21.5	23.3	24.6	24.2	22.4	20.4	16.8	16.5
	5%	DB	17.2	19.0	22.8	26.8	30.3	33.0	33.4	33.6	31.9	27.1	21.3	18.0
		MCWB	14.8	14.5	15.6	18.2	21.0	23.0	24.4	23.8	22.2	19.1	16.0	15.2
	10%	DB	15.1	16.8	20.7	24.9	28.7	31.7	32.1	32.1	30.1	25.2	19.4	15.9
		MCWB	12.4	12.9	15.0	17.3	20.4	22.8	24.1	23.6	21.7	18.4	15.1	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	19.3	19.9	21.8	24.0	25.9	26.8	26.6	24.9	23.8	20.7	19.0
		MCDB	20.3	21.8	23.5	27.2	28.9	31.3	32.1	32.4	30.5	27.2	23.2	21.5
	2%	WB	16.7	17.7	18.4	20.5	23.1	25.2	26.2	25.7	24.2	22.4	19.0	17.3
		MCDB	18.4	20.3	23.0	25.5	28.5	30.6	31.3	31.4	29.6	26.7	21.7	19.4
	5%	WB	15.1	15.7	17.1	19.6	22.4	24.6	25.6	25.2	23.6	21.1	17.5	15.6
		MCDB	17.1	18.0	21.2	24.3	27.9	30.0	30.8	30.7	28.7	25.0	20.3	17.6
	10%	WB	13.0	13.7	15.8	18.6	21.7	24.0	25.1	24.6	22.9	19.9	15.7	13.5
		MCDB	14.6	16.0	19.8	23.0	26.9	29.2	30.1	29.8	27.6	23.2	18.4	15.6
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.4	11.5	12.1	11.4	10.7	10.2	10.9	11.7	12.3	11.6	9.7
		MCDBR	10.9	12.9	13.9	13.8	13.0	12.6	11.9	12.9	13.5	13.9	13.7	11.1
		MCWBR	8.3	8.7	7.8	6.1	4.7	3.9	3.8	3.9	4.4	6.2	7.9	8.3
	5% WB	MCDBR	9.3	10.5	11.2	11.0	10.6	10.5	10.3	10.7	10.4	10.5	9.9	9.8
		MCWBR	8.0	8.4	7.4	5.7	4.6	4.2	4.1	4.1	4.1	5.6	7.0	8.4
Clear-Sky Solar Irradiance	taub	0.322	0.326	0.370	0.400	0.450	0.484	0.524	0.532	0.432	0.382	0.342	0.320	
	taud	2.479	2.465	2.367	2.276	2.181	2.120	2.029	1.993	2.261	2.383	2.439	2.503	
	Ebn at Noon	884	921	902	886	842	810	775	760	827	845	851	865	
	Edh at Noon	86	97	116	133	148	157	171	175	127	104	88	79	
All-Sky Solar Radiation	RadAvg	2.45	2.97	4.13	5.36	5.88	6.24	5.95	5.52	4.78	3.86	2.81	2.10	
	RadStd	0.19	0.37	0.39	0.37	0.46	0.49	0.33	0.30	0.42	0.50	0.30	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.73	N/A	N/A	+0.71	N/A	N/A	-62	-120	+204	+158
(47) Regional (32 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	+105

Nomenclature: See separate page